

Question_9:

Question description

Rathik organized technical round interview in Macrosoft for the set of computer science candidates.

The problem is to perform Implement a stack using single queue. you have to use queue data structure, the task is to implement stack using only given queue data structure.

Rathik have given the deadline of only 5 minutes to complete the problem.

Can you Help the candidates to complete the problem within the specified time limit ?

Function Description

x is the element to be pushed and s is stack

push(s, x)

- 1) Let size of q be s.
- 2) Enqueue x to q
- 3) One by one Dequeue s items from queue and enqueue them.

Removes an item from stack

pop(s)

- 1) Dequeue an item from q

Constraints

$0 < n, m < N$

$1 < arr[i] < 1000$

Input Format:

First line indicates n & m, where n is the number of elements to be pushed into stack and m is the number of pop operation need to be performed

next line indicates the n number stack elements

Output Format:

First line indicates top of the element of the stack

second line indicates the top of the element after the pop operation

✓ Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5 2
1 2 3 4 5
```

EXPECTED OUTPUT

```
top of element 5
top of element 3
```

Test Case 2

INPUT (STDIN)

```
15 4
9 8 7 6 5 5 6 7 8 3 11 2 3 4 5
```

EXPECTED OUTPUT

```
top of element 5
top of element 11
```

Code:

```
1  /*
2  Question_9:
3  Implement a stack using a single queue.
4  Push elements so the newest element stays at the front.
5  */
6
7  #include <iostream>
8  #include <queue>
9  using namespace std;
10
11  class Stack
12  {
13      queue<int> q;
14
15  public:
16      void push(int val)
17      {
18          int n = q.size();
19          q.push(val);
20          for(int i = 0; i < n; i++)
21          {
22              q.push(q.front());
23              q.pop();
24          }
25      }
26      void pop()
27      {
28          if(!q.empty())
29              q.pop();
30      }
31      int top()
32      {
33          return q.front();
34      }
35  };
36  int main()
37  {
38      int n, m;
```

```
39     cin >> n >> m;  
40     Stack s;  
41     int x;  
42  
43     for(int i = 0; i < n; i++)  
44     {  
45         cin >> x;  
46         s.push(x);  
47     }  
48     cout << "top of element " << s.top() << endl;  
49  
50     for(int i = 0; i < m; i++)  
51     |     s.pop();  
52     cout << "top of element " << s.top();  
53     return 0;  
54 }
```

Output:

Test Case_1:

```
5 2  
1 2 3 4 5  
top of element 5  
top of element 3  
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
15 4  
9 8 7 6 5 5 6 7 8 3 11 2 3 4 5  
top of element 5  
top of element 11  
PS C:\Users\Prabin Yadav\Document
```